

Notes: Fonseca & Liu (JF, 2024)

Mortgage Lock-In, Mobility, and Labor Reallocation

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Credit-record data	2
2.2	Mortgage rate delta	2
2.3	Moving and labor reallocation outcomes	3
2.4	Controls and fixed effects	3
3	Theoretical framework	3
3.1	Household decisions	3
3.2	Moving benefit	3
3.3	Net benefits	4
3.4	Decision rules	4
3.5	Predictions	4
4	Empirical specifications	4
4.1	Baseline moving regression	4
4.2	Instrumental variables design	5
4.3	Labor opportunity regression	5
4.4	Business entry and exit regression	5
5	Identification and empirical design	5
5.1	Main source of identification	5
5.2	Balancing tests	5
5.3	Asymmetry test	6
5.4	What the design cannot fully prove	6
6	Tables and figures	7
6.1	Figure 1 and Table I: mortgage rates and sample summary	7
6.2	Figure 2 and Table II: identification and baseline moving effects	7
6.3	Figure 3 and Figure 4: nonlinearity and labor opportunities	8
6.4	Table III: wage growth and moving rates by degree of lock-in	8
6.5	Table IV: business entry and exit	9
7	Short summary	9
8	Conclusion	10
8.1	Core conclusion	10
8.2	Economic intuition	10
8.3	Takeaway	10

1 Big picture

- **Question.** How do rising mortgage rates affect household mobility and labor reallocation when households have fixed-rate mortgages locked in at lower rates?
- **Core mechanism.** When current mortgage rates rise above a household’s locked-in rate, moving requires giving up a valuable low-rate mortgage and remortgaging at a higher rate. This raises the financial cost of moving.
- **Key variable.** The paper defines the mortgage rate delta, Δr , as the difference between the mortgage rate locked in at origination and the current mortgage rate. A lower or more negative Δr indicates stronger lock-in.
- **Main empirical challenge.** Households with different mortgage rates may differ systematically in credit quality, location, and moving propensity. The paper uses variation in mortgage-origination timing and aggregate rate movements to isolate plausibly exogenous variation in Δr .
- **Main result.** A 1 percentage point decline in Δr reduces moving by about 9% overall and about 16% during 2022–2024.
- **Asymmetry.** The relation between moving and Δr is asymmetric. When Δr is sufficiently positive, households can refinance without moving, so moving does not keep increasing with Δr .
- **Labor reallocation channel.** Mortgage lock-in reduces responsiveness to wage opportunities that require moving, measured by wage growth in surrounding counties 50 to 150 miles away.
- **Entrepreneurship channel.** Lock-in also dampens flows into and out of self-employment, consistent with lower mobility reducing job transitions and reallocative activity.
- **Economic takeaway.** Long-term fixed-rate mortgages insure households against interest-rate increases, but that insurance creates a mobility friction when rates rise.

2 Data and measurement

2.1 Credit-record data

- The core data are individual-level credit records from the Gies Consumer and Small Business Credit Panel.
- The sample covers millions of mortgage borrowers from 2010 to 2024.
- The data include mortgage origination date, mortgage balance, mortgage payment, remaining term, credit score, age, gender, county, ZIP code, and moving indicators.
- Moving is inferred from changes in address information in credit records.

Interpretation: The individual-level panel lets the authors observe mortgage contract terms, location, and moving behavior for the same household over time.

2.2 Mortgage rate delta

The main treatment variable is

$$\Delta r_{i,t} = r_i^{orig} - r_t^{current}, \quad (1)$$

where r_i^{orig} is the rate locked in at mortgage origination and $r_t^{current}$ is the prevailing market mortgage rate.

- If Δr is negative, the current market rate is higher than the locked-in rate, so moving and remortgaging is costly.
- If Δr is positive, the household can potentially benefit from remortgaging.

- The aggregate mortgage rate delta is based on the difference between current rates and the average origination-rate environment.

Interpretation: A decline in Δr means the household becomes more locked in.

2.3 Moving and labor reallocation outcomes

- **Moving rate:** indicator for any move.
- **Out-of-county move:** indicator for moves across county boundaries.
- **Business entry and exit:** transitions into and out of self-employment or small-business ownership.
- **Labor-opportunity shock:** wage growth in nearby counties within a 50- to 150-mile ring.

Interpretation: Out-of-county moves and wage-growth responses are closer to labor reallocation than local address changes.

2.4 Controls and fixed effects

Controls include mortgage balance, mortgage payment, remaining term, credit score, age, age squared, gender, and ZIP-code house price index. Main specifications also include ZIP-code fixed effects and county-by-year or origination-month-by-year fixed effects.

Interpretation: These controls absorb household financial characteristics, local housing-market conditions, and common time variation.

3 Theoretical framework

3.1 Household decisions

Households live for two periods and choose one of three actions:

$$D \in \{S, R, M\}, \quad (2)$$

where S means stay, R means refinance without moving, and M means move and remortgage.

The mortgage payment in period 2 is

$$M_2 = \begin{cases} r_1 L, & D = S, \\ r_2 L, & D \in \{R, M\}. \end{cases} \quad (3)$$

The period-2 interest rate is

$$r_2 = r_1 - \Delta, \quad \Delta r = r_1 - r_2 = \Delta. \quad (4)$$

Interpretation: Staying preserves the locked-in mortgage rate. Refinancing or moving exposes the household to the current rate.

3.2 Moving benefit

Households receive a moving opportunity η that raises income if they move:

$$Y_2 = \begin{cases} Y, & D \in \{S, R\}, \\ Y(1 + \eta), & D = M. \end{cases} \quad (5)$$

Moving costs κ_m and remortgaging costs κ_r .

Interpretation: The moving shock is interpreted empirically as an outside labor-market opportunity that requires relocation.

3.3 Net benefits

The net benefit of refinancing is

$$\Delta rL - \kappa_r. \quad (6)$$

The net benefit of moving and remortgaging relative to staying is

$$\eta Y + \Delta rL - \kappa_r - \kappa_m. \quad (7)$$

A household is a potential refinancer when

$$\Delta rL - \kappa_r \geq 0, \quad (8)$$

and a potential mover when

$$\eta Y - \kappa_m \geq 0. \quad (9)$$

Interpretation: Mortgage lock-in occurs when a household would move based on fundamentals, but the remortgaging penalty makes moving unattractive.

3.4 Decision rules

The household chooses to stay if

$$\Delta rL - \kappa_r < 0 \quad \text{and} \quad \eta Y + \Delta rL - \kappa_m - \kappa_r < 0. \quad (10)$$

It chooses to refinance if

$$\Delta rL - \kappa_r \geq 0 \quad \text{and} \quad \eta Y - \kappa_m < 0. \quad (11)$$

It chooses to move if

$$\eta Y - \kappa_m \geq 0 \quad \text{and} \quad \eta Y + \Delta rL - \kappa_m - \kappa_r \geq 0. \quad (12)$$

Interpretation: Refinancing breaks the symmetry: when rates are favorable, a household can capture the interest-rate gain without moving.

3.5 Predictions

- **Prediction 1:** Moving is increasing in Δr when households are locked in.
- **Prediction 2:** The moving- Δr relation flattens once $\Delta r \geq \kappa_r/L$.
- **Prediction 3:** Moving responsiveness to labor-market shocks is greater when households are less locked in.

Interpretation: These predictions map directly to the empirical tests: the paper estimates the effect of Δr on moving, tests asymmetry, and interacts lock-in with wage-growth shocks.

4 Empirical specifications

4.1 Baseline moving regression

The baseline relationship is

$$I(\text{Move}_{i,t}) = \alpha + \beta \Delta r_{i,t} + X'_{i,t} \gamma + \mu_z + \lambda_{c,t} + \varepsilon_{i,t}, \quad (13)$$

where μ_z are ZIP-code fixed effects and $\lambda_{c,t}$ are county-by-year or related time fixed effects.

Interpretation: The coefficient β measures how moving changes with the household-specific mortgage rate delta. A positive β means stronger lock-in, lower Δr , reduces moving.

4.2 Instrumental variables design

The first stage uses aggregate mortgage-rate movements:

$$\Delta r_{i,t} = \pi \text{Aggregate } \Delta r_t + X'_{i,t} \delta + FE + u_{i,t}. \quad (14)$$

The second stage is

$$I(\text{Move}_{i,t}) = \beta \widehat{\Delta r}_{i,t} + X'_{i,t} \gamma + FE + \varepsilon_{i,t}. \quad (15)$$

Interpretation: Origination timing creates variation in locked-in rates. Aggregate rate movements then generate differential lock-in across otherwise similar households.

4.3 Labor opportunity regression

For out-of-county moves, the paper estimates

$$I(\text{MoveOutCounty}_{i,t}) = \beta \text{WageGrowth}_{i,t} + X'_{i,t} \gamma + FE + \varepsilon_{i,t}, \quad (16)$$

separately for high- and low-lock-in groups. Wage growth in the 50- to 150-mile ring is instrumented with a shift-share instrument.

Interpretation: The test asks whether locked-in households are less responsive to nearby wage opportunities that require moving.

4.4 Business entry and exit regression

For entrepreneurship flows, the paper estimates

$$\text{BusinessOutcome}_i = \alpha + \beta \Delta r_{i,2019} + X'_i \gamma + FE + \varepsilon_i, \quad (17)$$

where the outcomes are business entry and business exit from 2020 to 2023.

Interpretation: If mortgage lock-in reduces mobility, it can also reduce transitions into or out of self-employment.

5 Identification and empirical design

5.1 Main source of identification

- Households lock in rates at different origination dates.
- Aggregate rates then move in ways that create variation in Δr across borrowers.
- Conditional on controls and fixed effects, origination timing is used to isolate plausibly exogenous exposure to rising rates.

Interpretation: The identifying assumption is that, after conditioning on observables and fixed effects, variation in origination timing is not driven by unobserved moving propensity.

5.2 Balancing tests

The paper runs balancing regressions showing whether borrower characteristics predict mortgage-rate deltas. With fixed effects and controls, observable imbalances are small.

Interpretation: The balancing evidence supports the credibility of the instrument and the fixed-effects design.

5.3 Asymmetry test

The theory predicts a kink: moving should respond to Δr when households are locked in, but should flatten when Δr is sufficiently positive because refinancing is available without moving.

Interpretation: The kink is important because it distinguishes mortgage lock-in from a simple linear effect of mortgage rates or borrower characteristics.

5.4 What the design cannot fully prove

- Credit-record data do not directly observe motives for moving.
- Wage-growth shocks are proxies for labor opportunities and may not match each household's job market.
- Business-entry and exit tests overlap with the pandemic period.
- The results estimate mobility and reallocation effects, not a complete welfare analysis of fixed-rate mortgages.

Interpretation: The empirical evidence is strongest for mobility responses and suggestive for broader labor reallocation effects.

6 Tables and figures

6.1 Figure 1 and Table I: mortgage rates and sample summary

Read:

- Figure 1 shows that 30-year fixed mortgage rates fell for decades, reached very low levels around 2020, and then rose sharply during 2022–2023.
- Table I reports 7.7 million observations in the baseline mortgage sample.
- The average moving rate is 7.52 percentage points, the average mortgage rate is 4.75 percentage points, and the average mortgage-rate delta is 0.53 percentage points.
- Average mortgage balance is about \$221,510 and average remaining mortgage term is about 21 years.

Economic message: The rate shock was large relative to borrowers’ locked-in rates, and the sample is large enough to identify mobility effects precisely.



Figure 1: average 30-year fixed mortgage rates

Table I
Summary Statistics

This table shows descriptive statistics for our 2010–2024 sample. Credit outcomes and demographics are from the Gies Consumer and Small Business Credit Panel. Average 30-year fixed mortgage rates are from the Federal Reserve Bank of St. Louis.

	Mean	Med.	St. Dev.
Moving rate (p.p.)	7.52	0.00	26.36
Out-of-county moving rate (p.p.)	4.40	0.00	20.50
Refinancing rate (p.p.)	5.44	0.00	22.68
Δr (p.p.)	0.53	0.40	2.23
Aggregate Δr	0.41	0.51	1.58
Mortgage rate (p.p.)	4.75	4.37	2.00
Average 30-year fixed mortgage rate (p.p.)	4.23	3.99	0.93
Mortgage balance (\$1,000)	221.51	165.29	233.13
Mortgage payment (\$1,000)	1.76	1.39	2.87
Remaining mortgage term (years)	21.21	24.00	7.88
Time since origination (years)	5.31	4.00	4.38
Credit score	754.70	781.00	81.34
Female (p.p.)	47.99	0.00	49.96
Age (years)	50.13	50.00	13.35
White collar occupation (p.p.)	44.14	0.00	49.66
Observations	7,707,003		

Table I: summary statistics

6.2 Figure 2 and Table II: identification and baseline moving effects

Read:

- Figure 2 shows balancing-regression coefficients for borrower characteristics.
- Once fixed effects are included, observable characteristics are largely balanced with respect to mortgage-rate deltas.
- Table II reports OLS, first-stage, reduced-form, and IV estimates of the effect of Δr on moving.
- The IV coefficient is about 0.69 for 2010–2024 and about 1.20 for 2022–2024.
- A 1 percentage point decline in Δr reduces moving by about 9% overall and about 16% in 2022–2024.

Economic message: The results support the lock-in mechanism: when households face a larger penalty from giving up their locked-in mortgage, they move less.

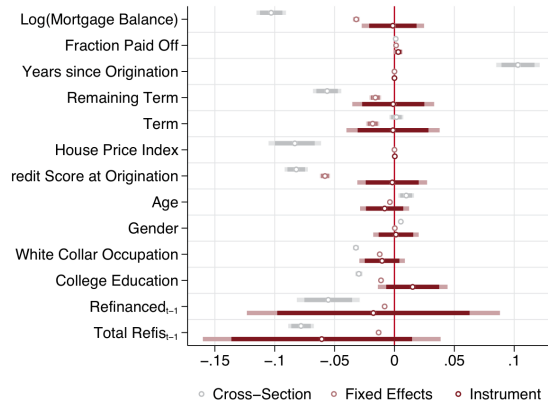


Table II
The Effect of Mortgage Rate Deltas on Moving Rates

Columns 1 and 5 report OLS estimates of equation (29). Columns 2 and 6 report estimates of the first-stage equation (27). Columns 3 and 7 report estimates of the reduced form equation (28). Columns 4 and 8 report two-stage least squares estimates of equation (29). In columns 1 to 4, we use our full sample from 2010 to 2024. In columns 5 to 8, we restrict attention to the period from 2022 onward. Controls include mortgage balance, mortgage payment, remaining term, credit score, age, age squared, gender, and a zip code house price index. Standard errors, reported in parentheses, are double clustered at the county and origination-month-year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Sample Period:	2010–2024				2022–2024			
	$I(Moved)$ OLS (1)	Δr FS (2)	$I(Moved)$ RF (3)	$I(Moved)$ IV (4)	$I(Moved)$ OLS (5)	Δr FS (6)	$I(Moved)$ RF (7)	$I(Moved)$ IV (8)
Δr	0.23*** (0.02)			0.66*** (0.07)	0.26*** (0.07)			1.20*** (0.25)
Aggregate Δr		0.57*** (0.01)	0.39*** (0.04)			0.66*** (0.02)	0.79*** (0.18)	
Zip code FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F-stat		3,169.63				1,755.95		
Observations	7,707,003	7,707,003	7,707,003	7,707,003	1,118,201	1,118,201	1,118,201	1,118,201

6.3 Figure 3 and Figure 4: nonlinearity and labor opportunities

Read:

- Figure 3 plots residualized moving rates against aggregate mortgage-rate deltas.
- Moving rates rise with Δr when Δr is low or negative, but the relation flattens at high positive Δr .
- Figure 4 shows that out-of-county moving responds more to nearby wage growth among households with high aggregate Δr .
- Households with lower Δr are less responsive to wage opportunities that require moving.

Economic message: The figures show both core predictions of the model: asymmetric moving responses and weaker labor reallocation when households are locked in.

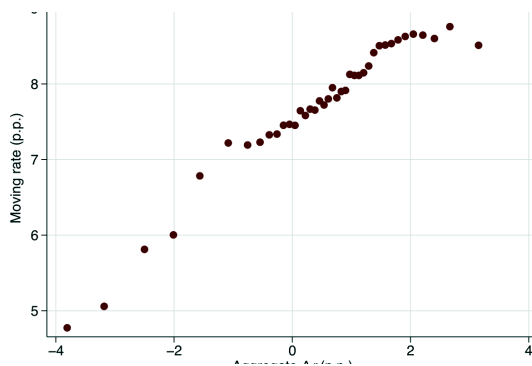


Figure 3: moving rates and aggregate Δr



Figure 4: moving rates and wage growth by lock-in

6.4 Table III: wage growth and moving rates by degree of lock-in

Read:

- Table III estimates the effect of out-of-county wage growth on out-of-county moving.
- For low aggregate Δr borrowers, the IV coefficient on wage growth is small and statistically insignificant.
- For high aggregate Δr borrowers, the IV coefficient is 0.20 and statistically significant.
- The high- and low-lock-in groups differ significantly in their response to wage opportunities.

Economic message: Mortgage lock-in dampens labor reallocation by making workers less responsive to better wage opportunities in nearby regions.

Table III
The Effect of Wage Growth on Moving Rates by Degree of Lock-In

Columns 1 and 4 report OLS estimates of the relationship between out-of-county moving rates and wage growth in counties within a 50- to 150-mile ring. Columns 2 and 5 report first-stage estimates of the Bartik wage growth IV. Columns 3 and 6 report two-stage least squares estimates of equation (30). High and low aggregate Δr refer to borrowers who are above or below the sample median aggregate Δr , respectively. Controls include mortgage balance, mortgage payment, remaining term, credit score, age, age squared, gender, and a zip code house price index. Standard errors, reported in parentheses, are double clustered at the county and origination-month-year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Dependent Variable:	<i>I[Moved out of County]</i>					
Aggregate Δr Group:	OLS (1)	Low FS (2)	IV (3)	OLS (4)	High FS (5)	IV (6)
Wage growth	-0.09*** (0.03)		-0.02 (0.09)	-0.01 (0.02)		0.20*** (0.06)
Wage growth IV		1.82*** (0.12)			1.77*** (0.09)	
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
F-stat		241.33			390.83	
P-value of (3) = (6)						0.04
Observations	3,342,173	3,342,173	3,342,173	3,813,962	3,813,962	3,813,962

6.5 Table IV: business entry and exit

Read:

- Table IV studies business entry and exit from 2020 to 2023 using Δr_{2019} .
- A higher 2019 mortgage-rate delta increases business entry and exit in the baseline mortgage-borrower sample.
- The placebo sample with no mortgage does not show the same positive relationship.
- This suggests that mortgage lock-in reduces entrepreneurship-related transitions, not simply that all households in high-rate locations behave differently.

Economic message: Lock-in reduces job and self-employment transitions, extending the mobility channel to labor-market reallocation and entrepreneurship.

Table IV
The Effect of Mortgage Rate Deltas on Business Entry and Exit

All columns report two-stage least squares estimates of equation (31). Columns 1 and 2 report results for our baseline sample of mortgage borrowers. Columns 3 and 4 report results for a sample of renters and outright owners. We construct a counterfactual Δr for each individual in the sample of renters and outright owners using the average prime rate in the year of their last observed move. Controls include mortgage balance, mortgage payment, remaining term, credit score, age, age squared, and gender in columns 1 and 2. In columns 3 and 4, we do not control for mortgage balance, mortgage payment, and remaining term since individuals do not have a mortgage. Standard errors, reported in parentheses, are double clustered at the county and origination-month-year level in columns 1 and 2 and at the county and year-of-last-move level in columns 3 and 4. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Sample:	Baseline		Placebo Check: No Mortgage	
Dependent Variable:	<i>Bus.</i> <i>Entry</i> _{2020→2023} (1)	<i>Bus.</i> <i>Exit</i> _{2020→2023} (2)	<i>Bus.</i> <i>Entry</i> _{2020→2023} (3)	<i>Bus.</i> <i>Exit</i> _{2020→2023} (4)
Δr_{2019}	0.26*** (0.06)	0.30*** (0.07)	-0.17 (0.16)	-0.07 (0.07)
Sample mean (p.p.)	5.72	2.02	1.84	0.48
% Change	4.63	14.87	-9.42	-15.34
Zip code FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	584,551	584,551	42,186	42,186

7 Short summary

- The paper studies how fixed-rate mortgage lock-in affects household mobility and labor reallocation.

- The key variable is Δr , the difference between the locked-in mortgage rate and current mortgage rates.
- A lower Δr means stronger lock-in because moving requires giving up a low-rate mortgage.
- A 1 percentage point decline in Δr reduces moving by about 9% overall and 16% during 2022–2024.
- The moving– Δr relation is asymmetric: it rises when households are locked in and flattens when refinancing becomes attractive.
- Locked-in households are less responsive to wage-growth opportunities that require out-of-county moves.
- Lock-in also dampens business entry and exit, suggesting broader labor reallocation effects.
- The results show a real side effect of long-term fixed-rate mortgages: they insure against rate increases but reduce mobility when rates rise.

8 Conclusion

8.1 Core conclusion

Fonseca and Liu show that mortgage lock-in materially reduces household mobility and labor reallocation. Rising mortgage rates increase the cost of moving for households with low fixed rates, and this effect is economically large during the 2022–2024 tightening cycle.

8.2 Economic intuition

A fixed-rate mortgage is valuable when market rates rise. Moving forces the household to prepay the old mortgage and borrow again at the new higher rate. This creates a wedge between the private benefit of moving and the financial cost of remortgaging. Some households who would otherwise move therefore stay put.

8.3 Takeaway

The paper highlights a trade-off in mortgage design. Long-term fixed-rate mortgages provide insurance against interest-rate risk, but they can slow mobility and labor reallocation when rates rise. Housing finance therefore affects not only housing markets, but also workers' geographic and occupational adjustment.